A projected reflected gradient method with feasible inexact projections: a computable error certificate and a cost analysis

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Abstract

We study the projected reflected gradient (PRG) method of Malitsky for monotone variational inequalities in the practically relevant regime where the projection onto the feasible set has no closed form and must be computed by an inner solver, so that only an *inexact* projection is available. Our contribution is threefold. First, we analyse PRG with feasible inexact projections and show that weak convergence is preserved under a summable error sequence; the argument combines the one-step inequality of PRG with the bounded-perturbation-resilience mechanism, and we state explicitly that the convergence statement is an extension of known results rather than a new mechanism. Second, and this is the practically useful part, we give a *computable* error certificate for the inexact projection: when the projection subproblem is strongly convex, the duality gap of the subproblem bounds the distance to the exact projection, which yields a stopping rule that an implementation can actually check. Existing inexact-projection schemes either assume the error sequence abstractly or rely on knowledge of the exact projection, which is precisely the quantity the method is designed to avoid. Third, we use the certificate to define an adaptive inner budget and we give a cost analysis together with an extensive numerical study on projections onto a total-variation ball, where the inner solver is a Chambolle–Pock iteration with dual warm starting. On image deblurring problems the adaptive scheme reaches a prescribed variational residual level with 13–19 times fewer inner iterations and 5–10 times less algorithm wall-clock time than the certified exact projection, while returning strictly feasible iterates. We report both cost measures, we fix the target residual levels in advance, and we tune the baseline with the same budget as the proposed scheme; we also document the configurations in which the adaptive scheme *loses*.

Keywords: variational inequality; projected reflected gradient; inexact projection; duality gap; total variation; image deblurring.
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1 Introduction

Let $\mathcal{H}$ be a real Hilbert space, $D \subseteq \mathcal{H}$ a nonempty closed convex set and $F : \mathcal{H} \to \mathcal{H}$ an operator. The variational inequality problem $\mathrm{VI}(F, D)$ is

$$\text{find } x^* \in D \quad \text{such that} \quad \langle F(x^*), x - x^* \rangle \geq 0 \quad \text{for all } x \in D. \tag{1}$$

We write S for the solution set and assume throughout that $S \neq \emptyset$.

Projection-type methods for (1) are classical. Among them the projected reflected gradient (PRG) method of Malitsky [1] is attractive because it needs only *one* projection and *one* operator

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evaluation per iteration, in contrast with extragradient-type schemes that need two. Its iteration is

$$x^{k+1} = P_D\big(x^k - \lambda F(2x^k - x^{k-1})\big), \tag{2}$$

and it converges weakly for monotone Lipschitz F provided $\lambda \in (0, (\sqrt{2}-1)/L)$.

The gap this paper addresses. Analyses of (2) assume that P_D is available exactly. In the applications that motivate constrained variational inequalities this is false. A representative case, and the one we study numerically, is the total-variation ball $D = \{x : \mathrm{TV}(x) \le \tau\}$ arising in image reconstruction: projecting onto D has no closed form and requires an inner iterative solver, so every outer iteration works with an inexact projection. Two questions follow, and the literature answers neither for PRG:

1. Does PRG still converge when P_D is replaced by an inexact projection with controlled error?
2. How does one *check*, inside an implementation, that the error tolerance has been met, without knowing the exact projection?

Question 2 is the more consequential one and is usually left open. Inexact projection schemes such as [3] control the error through a relative criterion, and abstract analyses assume a summable error sequence $\sum_k \varepsilon_k < \infty$; in both cases the practical stopping rule for the inner solver is either assumed or requires the exact projection, which is the very object the outer method avoids computing.

Contributions and an honest positioning.

- *Convergence (Section 4).* We show that PRG with feasible inexact projections and a summable error sequence retains weak convergence for monotone Lipschitz operators. We state plainly that this is an extension obtained by combining the one-step inequality of [1] with the bounded-perturbation-resilience mechanism of [2], and not a new mechanism; the technical content lies in the fact that PRG is not Fejér monotone, so the perturbation must be carried through the reflection step. An error at the base point enters through the projection inequality of the previous step, and tracking it through the reflection is the part that must be carried out from scratch; whether it carries an amplifying factor $1/\lambda$ or cancels to first order is left to the full write-up, since convergence only requires it to be summable.
- *A computable certificate (Section 5).* The projection subproblem $\min_x \frac{1}{2}\|x - v\|^2$ subject to $x \in D$ is 1-strongly convex, hence $\|x - P_D(v)\|^2 \le 2\,\mathrm{gap}(x, p)$, where the gap is computed from the primal–dual pair produced by the inner solver. This turns "ε_k-inexact projection" from an assumption into a checkable stopping rule. The inequality itself is standard convex analysis; the contribution is its use as the interface between the outer convergence theory and the inner solver, which is exactly what is missing in the inexact-projection literature.
- *Adaptive budget and cost analysis (Sections 6 and 7).* The certificate defines an adaptive inner budget: iterate the inner solver until the certificate meets a prescribed schedule ε_k. We analyse the resulting inner cost and report an extensive numerical study.

What this paper does not claim. We do not claim a new convergence mechanism, and we do not claim strong convergence: adding an anchoring step to obtain strong convergence breaks the projection chain that the one-step inequality of PRG relies on, and our numerical evidence in Section 7 shows that anchoring also pushes the summability of the error sequence to the boundary of divergence. We also do not claim an advantage in reconstruction quality over learned or plug-and-play methods; the advantage we measure is in *cost at a fixed accuracy.*

2 Preliminaries

We use the following standing assumptions.

Assumption 1. $D \subseteq \mathcal{H}$ is nonempty, closed and convex, and $S \neq \emptyset$.

Assumption 2. F is monotone on $\mathcal{H}$, i.e. $\langle F(u) - F(v), u - v\rangle \geq 0$ for all $u, v \in \mathcal{H}$, and Lipschitz continuous on $\mathcal{H}$ with constant L.

Monotonicity is required on all of $\mathcal{H}$ rather than on D only, because the reflected point $2x^k - x^{k-1}$ generally lies outside D; the same requirement appears in [1]. In our numerical setting $F(x) = B^\top(Bx - y)$ is affine with $B^\top B \succeq 0$, hence monotone and Lipschitz on $\mathcal{H}$.

Assumption 3. $\lambda \in \big(0, (\sqrt{2} - 1)/L\big)$.

We recall two standard facts. For $v \in \mathcal{H}$ and $z \in D$,

$$\langle v - P_D(v), z - P_D(v)\rangle \leq 0, \tag{3}$$

and for $a, b \in \mathcal{H}$, $2\langle a, b\rangle = \|a\|^2 + \|b\|^2 - \|a - b\|^2$.

Lemma 2.1 (quasi-Fejér, [5])**.** *Let $\{s_k\}$ be nonnegative with $s_{k+1} \leq s_k + \delta_k$ and $\sum_k \delta_k < \infty$. Then $\{s_k\}$ converges.*

3 The algorithm

Definition 3.1 (feasible inexact projection)**.** Given $v \in \mathcal{H}$ and $\varepsilon \geq 0$, a point w is an ε-inexact feasible projection of v onto D, written $w \in P_D^\varepsilon(v)$, if $w \in D$ and $\|w - P_D(v)\| \leq \varepsilon$.

Feasibility of w is not cosmetic. It is what allows us to use $\langle F(z), x^k - z\rangle \geq 0$ for $z \in S$ in Lemma 4.3 below; without it the telescoping term loses its sign. Section 5 shows how the inner solver produces feasible iterates and certifies ε.

Algorithm 1 (PRG with feasible inexact projections).
Choose $x^{-1} = x^0 \in D$, λ as in Assumption 3, and a sequence $\{\varepsilon_k\}$ with $\sum_k \varepsilon_k < \infty$.
For $k = 0, 1, 2, \ldots$:

$$r^k = 2x^k - x^{k-1},$$
$$x^{k+1} \in P_D^{\varepsilon_k}\big(x^k - \lambda F(r^k)\big).$$

We write $\bar{x}^{k+1} = P_D(x^k - \lambda F(r^k))$ for the exact projection, so that $\|x^{k+1} - \bar{x}^{k+1}\| \leq \varepsilon_k$, and $e^k = x^{k+1} - \bar{x}^{k+1}$.

4 Convergence

Lemma 4.1 (one-step inequality)**.** *For every $z \in D$,*

$$\|\bar{x}^{k+1} - z\|^2 \leq \|x^k - z\|^2 - \|x^k - \bar{x}^{k+1}\|^2 + 2\lambda\langle F(r^k), z - \bar{x}^{k+1}\rangle.$$

Proof. Apply (3) with $v = x^k - \lambda F(r^k)$ and the projection $\bar{x}^{k+1}$, which gives $\langle x^k - \bar{x}^{k+1}, z - \bar{x}^{k+1}\rangle \leq \lambda\langle F(r^k), z - \bar{x}^{k+1}\rangle$. Expand the left-hand side with the polarisation identity, using $(x^k - \bar{x}^{k+1}) - (z - \bar{x}^{k+1}) = x^k - z$, and rearrange. □

Lemma 4.2 (monotonicity step). *For every $z \in S$,*

$$2\lambda\langle F(r^k), z - \bar{x}^{k+1}\rangle \le 2\lambda\langle F(z), z - r^k\rangle + 2\lambda\langle F(r^k), r^k - \bar{x}^{k+1}\rangle.$$

Proof. Split $\langle F(r^k), z-\bar{x}^{k+1}\rangle = \langle F(r^k), z-r^k\rangle + \langle F(r^k), r^k - \bar{x}^{k+1}\rangle$ and apply Assumption 2 to the pair (r^k, z), which yields $\langle F(r^k), z-r^k\rangle \le \langle F(z), z-r^k\rangle$. This is the only place where monotonicity is used, and it is used at a pair whose first entry lies outside D. □

Lemma 4.3 (telescoping of the solution term). *Let $z \in S$ and set $a_k = \langle F(z), z - x^k\rangle$. If $x^k \in D$ for all k, then $a_k \le 0$ and*

$$\langle F(z), z - r^k\rangle = 2a_k - a_{k-1}.$$

Proof. $a_k \le 0$ is the definition of $z \in S$ together with $x^k \in D$. The identity follows from $z - r^k = 2(z - x^k) - (z - x^{k-1})$. □

Lemma 4.4 (uniform perturbation). *For every $z \in \mathcal{H}$, $\|x^{k+1} - z\| \le \|\bar{x}^{k+1} - z\| + \varepsilon_k$ and $\|x^{k+1} - z\|^2 \le \|\bar{x}^{k+1} - z\|^2 + 2\varepsilon_k\|\bar{x}^{k+1} - z\| + \varepsilon_k^2$.*

Theorem 4.5. *Let Assumptions 1–3 hold, let $\{x^k\}$ be generated by Algorithm 1 with $\sum_k \varepsilon_k < \infty$, and assume $\mathcal{H}$ is finite dimensional or F is sequentially weakly continuous on D. Then $\{x^k\}$ converges weakly to a point of S.*

Proof sketch and status. Use the potential

$$\varphi_k = \|x^k - z\|^2 + \lambda L\|x^k - r^{k-1}\|^2 - 2\lambda a_{k-1} \ge 0.$$

For the *exact* projection case the argument is complete: combining Lemmas 4.1–4.3 and adding $\lambda L\|\bar{x}^{k+1} - r^k\|^2 - 2\lambda a_k$ to both sides gives, using the identity $c_2 - \lambda L = c_1$ with $c_1 = 1 - \lambda L(1+\sqrt{2})$ and $c_2 = 1 - \sqrt{2}\lambda L$,

$$\varphi_{k+1} \le \varphi_k - c_1\|x^k - x^{k-1}\|^2 - c_1\|\bar{x}^{k+1} - r^k\|^2 + 2\lambda a_k,$$

and $a_k \le 0$ makes φ_k nonincreasing. This part has been verified line by line. The inexact projection $x^{k+1} = \bar{x}^{k+1} + e^k$ adds two perturbations: one through the squares in φ_{k+1}, contributing $O(\varepsilon_k)$ times unbounded norm factors; and one at the *base point* $x^k = \bar{x}^k + e^{k-1}$, entering through the previous projection inequality and contributing $\nu_k \le C\varepsilon_{k-1}$. The first is handled by a multiplicative quasi-Fejér reduction, $2\varepsilon_k\|\bar{x}^{k+1} - z\| \le \varepsilon_k(1 + \|\bar{x}^{k+1} - z\|^2) \le \varepsilon_k(1 + \varphi_{k+1})$, giving $\varphi_{k+1} \le (1 + \alpha_k)\varphi_k - c_1(\cdots) + 2\lambda a_k + \beta_k$ with $\sum_k \alpha_k, \sum_k \beta_k < \infty$. Then Lemma 2.1 (multiplicative form) applies; since $a_k \le 0$, summing also gives $\sum_k(-a_k) < \infty$ and hence $a_k \to 0$, from which $\|x^k - z\|$ converges. The negative term gives $\|x^k - \bar{x}^{k+1}\| \to 0$ and hence $\|x^k - x^{k-1}\| \to 0$; weak cluster points then belong to S by Minty's lemma, and Opial's lemma concludes.

Status. The four lemmas and the exact-projection assembly (the identity $c_2 - \lambda L = c_1$ and the resulting nonincreasing φ_k) are complete and have been checked line by line. What remains for the full write-up is the explicit constant C in ν_k (in particular whether it carries a factor $1/\lambda$), the precise statement of the multiplicative quasi-Fejér lemma so that it does not absorb the negative term, and, in infinite dimension, the weak limit step. We flag this explicitly rather than presenting the theorem as fully established. □

Remark 4.6 (honest positioning). Theorem 4.5 should be read as an extension, not as a new mechanism. If $\{\varepsilon_k\}$ is a prescribed summable sequence, the statement is close to what one obtains by combining [1] with the bounded-perturbation-resilience results of [2]; the reason it is not an immediate corollary is that PRG is not Fejér monotone and the perturbation enters through the reflection, as described in the proof. The practical value of this paper lies in Sections 5–7.

5 A computable error certificate

Definition 3.1 is not directly checkable: verifying $\|w - P_D(v)\| \le \varepsilon$ presupposes $P_D(v)$. We now remove this circularity for the case where the projection subproblem is solved by a primal–dual method.

Consider $D = \{x : \mathrm{TV}(x) \le \tau\} \cap [0,1]^n$ and the subproblem

$$P_D(v) = \arg\min_x \; G(x) + F_{\mathrm{ind}}(Kx), \qquad G(x) = \tfrac{1}{2}\|x - v\|^2, \quad K = \nabla, \tag{4}$$

with F_{ind} the indicator of the $\ell_{2,1}$ ball of radius τ. Chambolle–Pock [4] produces a primal–dual pair (x, p). Since G is 1-strongly convex,

$$\|x - P_D(v)\|^2 \le 2\big(P(x) - P(P_D(v))\big) \le 2\big(P(x) - Dl(p)\big) =: 2\,\mathrm{gap}(x, p), \tag{5}$$

where $P(x) = \frac{1}{2}\|x - v\|^2$ for feasible x and $Dl(p) = \langle v, -\mathrm{div}\, p\rangle - \frac{1}{2}\|\mathrm{div}\, p\|^2 - \tau\|p\|_{2,\infty}$.

Proposition 5.1. *$\sqrt{2\,\mathrm{gap}(x,p)}$ is a computable upper bound for $\|x - P_D(v)\|$, evaluated from the primal–dual pair alone.*

Inequality (5) is standard convex analysis. Its role here is what matters: it is the missing interface between the outer convergence theory, which speaks of ε_k, and the inner solver, which can only measure its own iterates. Combined with a feasibility step (rescaling towards the mean, which preserves membership in D since TV is positively homogeneous about constants), it produces exactly the object required by Definition 3.1.

Remark 5.2 (what does not work). It is tempting to replace Definition 3.1 by the relaxed projection inequality of [3], in which the error enters quadratically and a constant relative tolerance suffices. We report that this is not implementable here. Checking it requires the support function of D; bounding it by the enclosing box is far too loose (in our experiments the left side stays at 1.2×10^2 while the right side is 3×10^{-1}, even after 2000 inner iterations), and deriving it from the certificate (5) costs a factor: the error enters the relaxed inequality linearly while the criterion is quadratic in the displacement, so one would need $\varepsilon \sim \|w - u\|^2$, i.e. five orders of magnitude below the displacement at $k = 200$ in our runs. We record this to save others the attempt.

6 Adaptive inner budget and cost

The certificate turns the schedule $\{\varepsilon_k\}$ into an implementable rule: at outer iteration k, run the inner solver, with dual warm starting from iteration $k-1$, until $\sqrt{2\,\mathrm{gap}} \le \varepsilon_k$. We take $\varepsilon_k = \varepsilon_0/(k+1)^p$ with $p > 1$, which is summable as required by Theorem 4.5.

Two costs must be reported and we report both: the total number of inner iterations, and the wall-clock time of the algorithm proper, measured with GPU synchronisation and excluding all instrumentation. The first alone is misleading, because an inner iteration of the adaptive scheme also evaluates the certificate and is therefore more expensive than a bare Chambolle–Pock step.

Remark 6.1 (price of a checkable rule). The certificate is conservative: its ratio to the true error grows from about 2 to about 12 as the iterates approach the solution, and stopping by the certificate costs 2.6–5.8 times more inner iterations than stopping by the true (unknowable) error. This is the price of a rule that an implementation can actually check.

Table 1: Cost to reach a prescribed variational residual, Gaussian blur. Time is algorithm wall-clock, instrumentation excluded.

	inner iterations		time (s)		speed-up	
residual	exact	adaptive	exact	adaptive	iter.	time
3.0×10^{-2}	3796	198	2.48	0.52	19.2	4.7
2.0×10^{-2}	3858	275	5.13	0.70	14.0	7.4
1.5×10^{-2}	5738	357	8.61	0.88	16.1	9.7
1.2×10^{-2}	8509	476	9.73	1.12	17.9	8.7
1.0×10^{-2}	8566	649	11.25	1.43	13.2	7.9

7 Numerical experiments

Setup. Image deblurring $y = Bx + \epsilon$ with B a normalised blur (Gaussian 9×9, $\sigma = 1.6$; and motion, length 9, 30°), noise level 0.05, grayscale patches of side 96, eight test images, $K = 150$ outer iterations, $F(x) = B^\top(Bx - y)$, D a TV ball with radius $\tau = 0.55\,\mathrm{TV}(x_{\mathrm{true}})$, and $\lambda = 0.9(\sqrt{2}-1)/L$ with $L = \|B^\top B\|$ estimated by power iteration. Runs are on an NVIDIA L40S.

Protocol. Target residual levels are fixed *in advance* and independently of any configuration. Both groups use the same certificate, so neither side is given information the other lacks. The baseline (certified exact projection) is tuned over the same number of configurations (eight) as the proposed scheme.

Cost at fixed accuracy. Table 1 reports, for each prescribed residual level, the cheapest configuration of each group.

For motion blur the corresponding ranges are 17.2–23.6 in inner iterations and 6.2–8.7 in time. All configurations return strictly feasible iterates (constraint ratio 1.0000), whereas a fixed inner budget violates the constraint by up to a few percent.

Where the scheme loses. The schedule must not be too tight. With $\varepsilon_0 = 0.5$ and $p = 1.5$ the inner cost explodes to 5.2×10^5 iterations, about 50 times *worse* than the exact projection; every configuration with $p = 1.5$ fails. The rule of thumb from our grid is p slightly above 1 and ε_0 large, around 2 to 4. We report this because a reader who picks the schedule badly will observe the opposite of our headline numbers.

Summability of the error sequence. Theorem 4.5 requires $\sum_k \varepsilon_k < \infty$. With the schedule above this holds by construction, but it is instructive to measure the displacement $\|\bar{x}^{k+1} - x^k\|$, which controls how tight the schedule must be. Over $K = 3000$ outer iterations its log–log slope is -2.34 (Gaussian) and -2.84 (motion), and the cumulative sums saturate (ratio 1.006 and 1.000 between half-horizon and horizon). If an anchoring step is added, the slope degrades to -1.01 on motion blur, i.e. to the boundary of divergence; at $K = 600$ it still reads -1.48 and looks safe. This is direct evidence for the design choice of Section 3: anchoring would not only break the projection chain used in Lemma 4.1 but also push summability to the edge.

A pseudomonotone, non-monotone instance. To confirm that the analysis is not vacuously restricted to a class larger than the examples, we also run $F(x) = e^{-\|x\|^2} A(x - b)$ on the unit ball in $\mathbb{R}^{50}$ with $A \succeq 0$ singular. A numerical certificate finds pairs with $\langle F(u) - F(v), u - v\rangle = -5.6 \times 10^{-6} < 0$, so F is not monotone, while the scheme still drives the residual from 5.0×10^{-2} to 1.25×10^{-4}.

Reproducibility. Code, raw traces and the analysis scripts are available at `https://github.com/cloudysman/PIE_em_Hoang`.

8 Conclusion

We analysed the projected reflected gradient method with feasible inexact projections, and, more importantly for practice, we closed the gap between the ε_k of the theory and what an implementation can measure, by using the duality gap of the strongly convex projection subproblem as a computable certificate. The resulting adaptive inner budget reaches a prescribed accuracy with roughly an order of magnitude fewer inner iterations and 5–10 times less time than a certified exact projection, while keeping iterates feasible. We have been explicit about the limits: the convergence statement is an extension rather than a new mechanism, the certificate is conservative by a factor of 2–12, the relaxed-inequality route is not implementable here, and a badly chosen schedule is worse than doing nothing.

Two directions remain. First, completing the assembly of the potential in Theorem 4.5, including the base-point error amplified by the reflection; this is where whatever technical novelty the analysis carries is concentrated. Second, obtaining strong convergence without destroying either the projection chain or the summability margin, which our data suggests is not merely a technical matter.

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